

## Objective

You are required to build an AI agent with a Local File Search MCP server that can find PDF files stored locally in a folder on a user's machine.

There is no strict requirement for the MCP server to search inside PDF content using full-text search or semantic search. However, implementing this capability would be considered a strong enhancement.

At a minimum, the Local File Search MCP server must support searching by file metadata, including:

- File name
- Folder path
- File extension
- Created date
- Modified date
- File size

## File Requirements

1. You may create any number of files for testing.
2. The test data must include at least 3–4 different file extensions, such as .docx, .xls, .pdf, .jpg, and .txt.
3. None of the file metadata or content should be related to Microsoft, IT, AI, or technology.
4. Choose a non-technology topic for the sample files. Suggested topics include Zoology, Biology, or another natural science domain.

## Remote MCP Requirement

The agent must also use an additional remotely hosted MCP server: the Microsoft Learn MCP Server.

Remote MCP endpoint:

<https://learn.microsoft.com/api/mcp>

Reference:

<https://learn.microsoft.com/en-us/training/support/mcp>

## Expected Final Product

The final solution should include:

- An OpenAI GPT-5.x LLM-based agent with two MCP/tools attached:

- Local File Search MCP server
  - Remote Microsoft Learn MCP server
- The Local File Search MCP server does not need to be hosted externally. An in-process implementation is acceptable.
- Prompting should follow the SKILLS.md style.
- A web-based UX is not required. A CLI-based interface is acceptable.
- When the implementation is ready to test, reach out for the OpenAI API key.
- The completed code must be checked into a GitHub repository.
  - The repository may be public, or
  - It may be shared with the GitHub handle abin-aot.
- The solution must use a code-first agentic AI framework, such as LangChain, Microsoft Agent Framework, or another industry-standard framework.

## Response Rules

1. Queries related to files on the local system must return only a strongly typed object or JSON response.
  - No additional textual explanation should be included.
2. Queries related to Microsoft technologies or Azure must be answered using the Microsoft Learn MCP server.
3. Microsoft technology or Azure responses must be precise and must not exceed 2,000 characters.
4. General-topic responses are not required.
  - The agent should only respond to:
    - Microsoft technology or Azure-related queries
    - Local file search queries

## Sample Queries

### Sample Query 1

User query:

“What PDF files are available in our system?”

**Expected behavior:**

The agent should use the Local File Search MCP server and return a JSON response listing all PDF files.

### Sample Query 2

User query:

“What is Azure Blob Storage?”

**Expected behavior:**

The agent should use the Microsoft Learn MCP server and provide a precise explanation of Azure Blob Storage.